



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BASIC SURVEYING PROJECT PLANNING

LSVPP401

Carry Out Basic Surveying Project Planning

Competence

RQF Level: 4

Learning Hours



60

Credits: 6

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: General

Curriculum: CBSLSV4001 –TVET certificate IV in land surveying

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Purpose statement	This module describes the knowledge, skills, and attitudes required to carry out basic surveying project planning. It is designed for students pursuing RQF level 4 in land surveying. At the end of this module, the students will be able to develop a risk management plan, create a project scheduling plan, elaborate cost management plan, and elaborate monitoring and evaluation plan.					
Learning assumed to be in place	Surveying project quantities and cost estimation, applied mathematics, technical drawing interpretation, surveying technical report, computer literacy.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group work and presentation	30%				
	• Individual work	40%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Develop risks management plan	1.1 Risks associated to a surveying project are clearly analysed in line with the project scope.
	1.2 Risk response plans are properly developed based on the analysis done.
	1.3 Risks are properly monitored and checked to ensure a continuous risk management.
2. Create a Project scheduling plan	2.1. Work structure is efficiently broken down into small work packages in line with the scope of the project.
	2.2. Tasks relationship is appropriately identified according to their types.
	2.3. Resources are appropriately assigned for each work activity according to their requirements.
	2.4. Duration is accurately estimated for each work activities with

	reference to their scope.
	2.5. Activities relationships are accurately represented according to the schedule of activities.
3. Elaborate Cost management plan	3.1. Project resource plan is appropriately elaborated with reference to the work breakdown structure.
	3.2. Cost estimation is accurately calculated in line with the estimation techniques used.
	3.3. Project budget is accurately estimated with reference to the project resources.
	3.4. Cost control plan is suitably established according to the project schedule and budget.
4. Elaborate monitoring and Evaluation plan	4.1. Monitoring tools are appropriately developed according to the project scheduling plan.
	4.2. Monitoring plan is properly developed in line with time, cost, risks and scope of the project.
	4.3. Evaluation tools are appropriately developed according to the project scheduling plan.
	4.4. Evaluation plan is properly developed in line with time, cost, risks and scope of the project.

Intended Knowledge, Skills, and Attitudes

Knowledge	Skills	Attitudes
✓ Identify risks ✓ Develop risk response plans and strategies ✓ Identify surveying projects tasks ✓ Sequence tasks in surveying projects ✓ Develop work breakdown structure ✓ Identify monitoring tools ✓ Identify evaluation tools ✓ Develop a scheduling tool	✓ Monitor risks ✓ Estimate activity costs ✓ Estimate activity durations ✓ Perform project monitoring ✓ Perform project evaluation ✓ Draw project tasks network diagrams ✓ Control project costs ✓ Estimate project resources ✓ Produce Gantt chart	✓ Use multitasking in surveying project resources estimation ✓ Use creativity and innovation during the developing of the monitoring and evaluation plans ✓ Pay attention to project cost details ✓ Demonstrate punctuality

		during the implementation of surveying projects ✓ Demonstrate resourcefulness in surveying project planning
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Course content

Learning outcomes	At the end of this module the learner will be able to: 1. Develop risks management plan 2. Elaborate cost management plan 3. Elaborate monitoring and evaluation plan 4. Create a project scheduling plan
Learning outcome 1: Develop risks management plan	Learning hours: 13
Indicative content	
• Risk analysis ✓ Hazard identification ✓ Project risk identification ✓ Risk assessment ✓ Risk analysis methods ✚ Qualitative risk analysis ✚ Quantitative risk analysis • Development of risk response plan ✓ Risk response planning process ✓ Risk response strategies ✓ Components of risk response plan • Risk monitoring and checking	

- ✓ Purpose of risk monitoring
- ✓ Tools and techniques for risk monitoring control
- ✓ Control inputs for monitoring risks
- ✓ Control outputs for monitoring risks

Resources required for the learning outcome

Equipment	Computer, Projector, white/blackboard, Printer.
Materials	Chalk, notebooks, paper, flip chart.
Tools	Marker pens, pencils.
Facilitation techniques	Individual and group work, Group discussion, Practical exercise, Trainer guided.
Formative assessment methods /(CAT)	Written assessment, Oral assessment, Product based assessment.

Learning outcome 2: Create a Project scheduling plan

Learning hours: 22

Indicative content

- **Work breakdown structure**
 - ✓ Analysis of project documents
 - ✓ Identification of work structure elements
 - ✓ Work structure elements decomposition.
 - ✓ Creating WBS dictionary
 - ✓ Methods of Work breakdown structure
 -  Deliverable-based method
 -  Phase-based method
- **Identification of the tasks relationship**
 - ✓ Types of task relationship

 Finish to Start

 Finish to Finish

 Start to Start

 Start to Finish.

✓ Identification of sequences of activities

 Activity on Arrow(AOA) representation

 Activity on node (AON) representation

- **Assigning resources to work activities**

✓ Labour

✓ Equipment

✓ Material

- **Estimation of work activities duration**

✓ Methods

 Analogous estimating

 Parametric estimating

 Three point estimating

 Heuristic estimating

 Reserve analysis

✓ Calculating activity durations

- **Presentation of activities relationship**

✓ Gantt chart

✓ Critical path method or Network analysis system

Resources required for the learning outcome

Equipment	Computer, white/blackboard, projector, printer.
Materials	Marker pen, chalk, paper.
Tools	Calculator, pens, pencils.
Facilitation techniques	Practical works, Group discussion, Trainer guided, Individual and group work
Formative assessment methods /(CAT)	Written assessment, Product based assessment, Performing assessment

Learning outcome 3: Elaborate Cost management plan

Learning hours: 15

Indicative content

- **Elaboration of project resource plan**
 - ✓ Project resources
 -  Human
 -  Financial
 -  Equipment
 - ✓ Resources scheduling
- **Cost estimation techniques**
 - ✓ Methods
 -  Rough/Approximate estimate
 -  Detailed estimate
 - ✓ Estimation of quantity
 - ✓ Estimation of unit price
 - ✓ Estimation of total cost
- **Estimation of cost control plan**
 - ✓ Identification of methods used in budget estimation
 -  Analogous
 -  Parametric
 -  Top-down
 -  Bottom-up
 -  3-point estimation
 - ✓ Estimation of direct cost
 - ✓ Estimation of overhead cost
 - ✓ Calculation of tax cost
 - ✓ Estimation of risk and contingencies cost

- ✓ Calculate profit

- **Establishment of cost control plan**

- ✓ Components of cost control plan
- ✓ Techniques of cost control
 - ✚ Continuous Key Performance Indicator Tracking
 - ✚ Consider inflation
 - ✚ Time tracking and management
 - ✚ Profitability tracking
 - ✚ Review budget if necessary
- ✓ Procedures

Resources required for the learning outcome

Equipment	Computer, white/blackboard, projector, printer.
Materials	Chalk, notebooks, paper, marker pen.
Tools	Calculator, pens, pencils.
Facilitation techniques	Practical works, Group discussion, Trainer guided, Individual and group work
Formative assessment methods /(CAT)	Written assessment, Product based assessment, Performing assessment

Learning outcome 4: Elaborate monitoring and evaluation plan

Learning hours: 10

Indicative content

- **Developing monitoring tools**
 - ✓ Elements of project monitoring plan
 - ✚ Time
 - ✚ Cost
 - ✚ Scope
 - ✚ Quality

- ✚ Risk
- ✚ Human resources
- ✚ Asset version
- ✚ Productivity
- ✚ Additional measures

✓ Project management tools

- ✚ Gantt chart
- ✚ Critical path
- ✚ Work breakdown structure(WBS)
- ✚ Project network diagram
- ✚ Project report
- ✚ Project management software
- ✚ Time sheet,
- ✚ Checklists

✓ Advantages of using tool for monitoring project

• **Developing of monitoring plan**

- ✓ Phases of project monitoring plan
- ✓ Steps in developing monitoring plan

• **Developing evaluation project tools**

✓ Project evaluation tools

- ✚ Questionnaires
- ✚ Interviews
- ✚ Focus groups
- ✚ Expert panel
- ✚ SWOT analysis
- ✚ Multi-criteria analysis
- ✚ Cost effectiveness analysis
- ✚ Types of evaluations
- ✚ Self-evaluation

-  Internal evaluation
-  Independent evaluation
-  External evaluation

✓ Phases and process of project evaluation

-  Planning phase
-  Implementation phase
-  Completion phase
-  Dissemination and reporting

• **Developing evaluation plan**

✓ Elements of evaluation plan

-  Purpose of the evaluation
-  Evaluation questions
-  Evaluation criteria
-  Timetable and work plan
-  Collecting data for an evaluation
-  Data collection methods to answer evaluation questions
-  Data collection tools and activities
-  Data analysis.

✓ Steps of developing evaluation plan

Resources required for the learning outcome

Equipment	Projector, Computer, white/blackboard, printer.
Materials	Paper, notebooks, marker pen.
Tools	Calculator, pens, pencils.
Facilitation techniques	Individual and group work, Trainer guided, Group discussion, Practical works.
Formative assessment methods /(CAT)	Written assessment, Performing assessment, Product based assessment.

References

1. Holmes, G. (2015). *Planning, Scheduling*, . LONDON: ReineCatch .
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4. MANZOOR, A. (2019). *Project Cost*. United States of America: Amir Manzoor.
5. Rumane, A. R. (2017). *Construction management*. BOCA RATON: TAYRLOR.